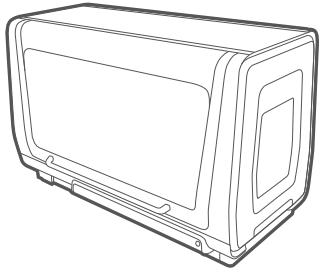
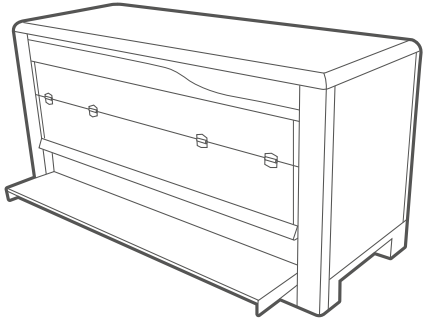
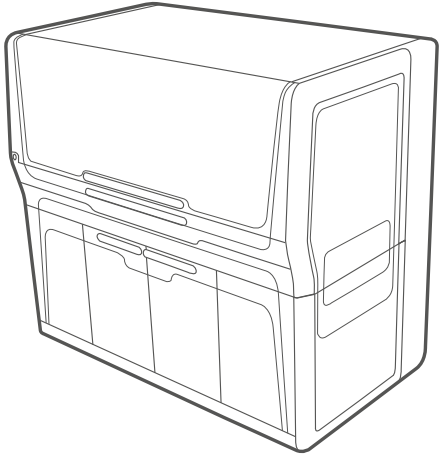
Hamilton Robotics Good Practice Guide







624727/00

HAMILTON

Document history

DOCUMENT TITLE	Good Practice Guide
VERSION	1.00
PART NO:	624727/00

Version	Revision	Issue	
1	00	2014-04-23	New edition

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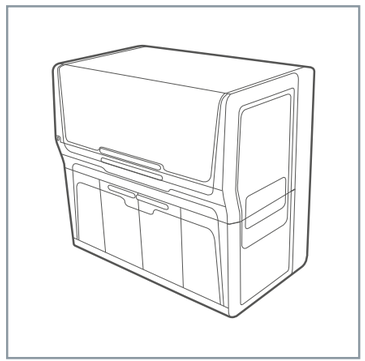
About this document

For your Safety
Read the respective instrument Operator's Manual before performing any work on the Hamilton instrument.

About this document
This document provides good practice guidelines for proper use and safe operation of Hamilton instruments.

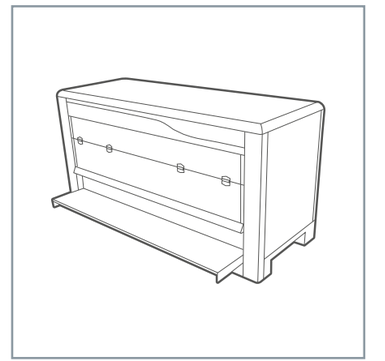
Scope
This document applies to the following Hamilton Instrument Lines:
- Hamilton VANTAGE Liquid Handing System
- Microlab STAR
- Microlab NIMBUS

Disclaimer
This document has been prepared to inform and guide laboratory employees working with a Hamilton instrument. It is intended to cover the good practice handling and safety aspects most often encountered in day-to-day use of the instrument.
The information contained in this manual is subject to change without notice. Hamilton Company does not assume any responsibility for the completeness of illustrations, explanations, instructions or statements provided in this document.



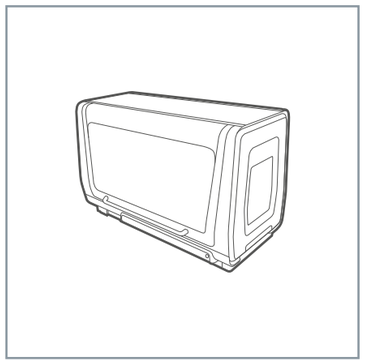
Hamilton VANTAGE

The Hamilton VANTAGE Liquid Handling System is a modular, next-generation platform which can be expanded as application and integration needs change.



Microlab STAR

The Microlab STAR is a flexible desktop pipetting system for simple to complex application needs.



Microlab NIMBUS

The Microlab NIMBUS is a desktop liquid handling instrument for low- to medium-throughput labs.

Safety

Instrument-related hazards

The safety of personnel and equipment can only be ensured if the safety instructions and the safety-related warnings are strictly observed and followed.
The following section describes the main safety considerations in operating your Hamilton product and the main hazards involved.



General Warning Symbol

- The general warning symbol indicates possible injury and damage to equipment if the instructions are not followed.



Electrical Hazard Symbol

- The electrical hazard symbol indicates possible injury and damage to equipment if the instructions are not followed.



Pinch Point, Mechanical Hazard Symbol

- The pinch point symbol indicates possible injury and damage to equipment by mechanical parts.



Laser Hazard Symbol

- The laser hazard symbol indicates possible injury if staring into the laser beam.



Hot Surface Symbol

- The hot surface hazard symbol indicates possible injury if touching heated devices or surfaces.



UV Light Hazard

- The UV light hazard symbol indicates possible injury from UV radiation. Consult the Operator's Manual, before switching on the UV-Light



Magnetic Hazard

- The magnetic hazard symbol indicates possible interference with cardiac pace makers or other personal devices.



Biological Hazard

- The biological hazard symbol indicates potentially contaminated parts of the instrument associated with the samples parts of theed.

Safety measures

Access to Power Switch/ Power Cord
Ensure that there is enough room to access the power switch and the power cord.

Working Area
The safety concept assumes that the doors are always closed and locked during normal operation.

Surrounding Area
Make sure the ventilation outlets of the instrument is not impaired by obstacles placed in the surrounding area.



CAUTION
Rapid temperature changes or direct sunlight may affect certain functions such as barcode reading and pipetting accuracy.

- Do not open windows next to the instrument
- Do not expose the instrument to direct sunlight or intense artificial light during operation

Substance-related hazards



WARNING

Chemical, biological and radioactive hazards can be associated with the substances used or the samples processed with the instrument. The same applies to waste disposal. The handling of substances and the disposal of waste may be subject to local, state or federal law or regulations with regard to health, environment or safety.

- Always be aware of possible hazards associated with these substances
- Use appropriate protective clothing, goggles and gloves
- Strictly observe the corresponding provisions



WARNING

Caustic substances can cause burns and eye injury.

- Always be aware of possible hazards associated with these substances
- Avoid exposure to caustic substances
- Use appropriate protective clothing, goggles and gloves



WARNING

Fire hazard.

- Use caution when using flammable materials. The instrument is not explosion protected nor for use in Ex zones.

Safety measures

Instrument parts and solid waste may be contaminated with potentially infectious substances.



Protective clothing, goggles and gloves.

Use appropriate:



protective clothing,



goggles



and gloves.

Decontamination

For your Safety

Apart from regular decontamination, the user must thoroughly decontaminate the instrument according to local regulations in the following cases:

- Before any maintenance or service work is performed on the instrument
- In case of accidents (e.g. crash, spilt substances, etc.)
- Before the instrument or parts of it is returned to HAMILTON (e.g. for repair)
- Prior to storage of the instrument
- Prior to disposal of the instrument or parts of it
- Generally before the instrument or parts of it leave the user's site

Method

Consult the designated instrument manual for decontamination procedures.

To decontaminate the instrument, perform the Weekly Maintenance.

Decontaminating

Use Microcide SQ, Deconex or Solarsept instead of water to clean the stop disks, o-rings, and tip eject sleeves.

General conduct of safety

Legal Regulations

All local, state and federal laws which prescribe the use, application, and/or the handling of dangerous materials in connection with the instrument must be strictly followed.

Duty of Maintenance and care

The decontamination method must be adapted to the respective application and the substances associated with it. The user takes the full responsibility for the appropriate decontamination of the entire equipment.

Spare Parts to be used

Make sure to only use consumables and spare parts recommended by HAMILTON.

Only HAMILTON Consumables

For best pipetting performance, only use original Hamilton tips and needles.



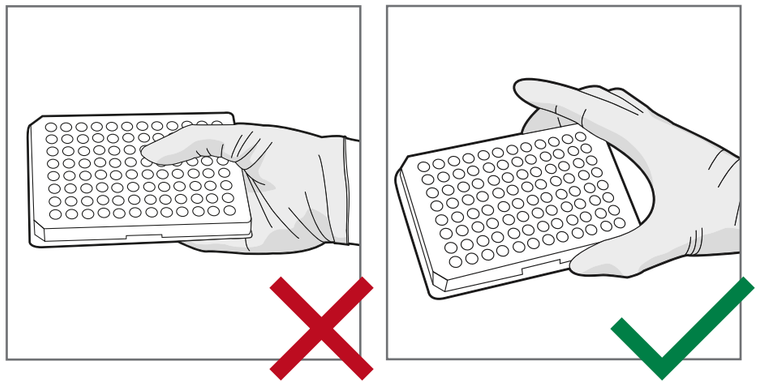
The instrument loading tray is a part of the instrument and must be threated with care.

CAUTION

Risk of damage to equipment. Leaning onto the instrument loading tray will lead to damage of the loading tray.

- Do not lean onto the loading tray

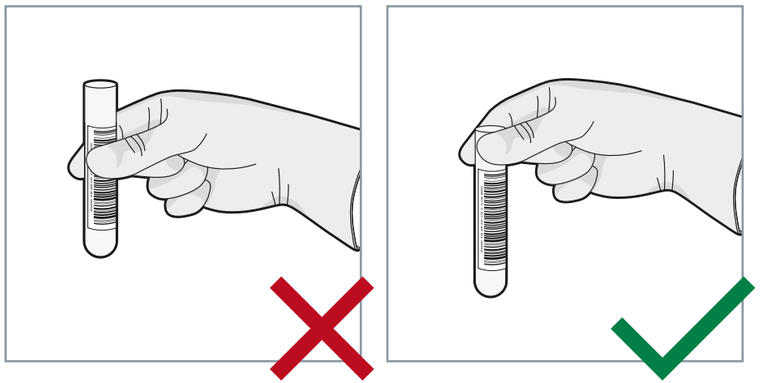
Holding labware



Microtiter plate

Hold Microtiter Plate by its side faces.

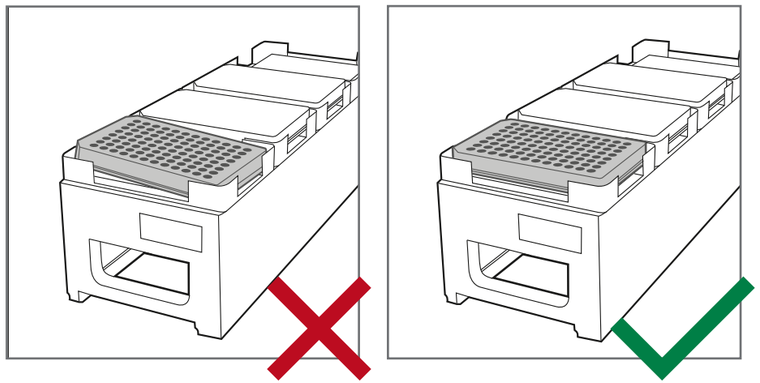
Holding tubes



Holding tubes

Avoid holding tubes on barcodes.
Hold the tubes as shown.

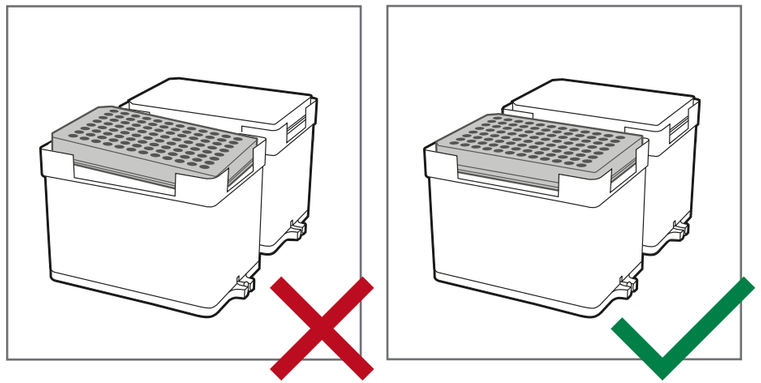
Placing labware on carriers and pedestals



Placing labware

Make sure that labware is placed onto the carrier correctly.

Incorrectly placed labware will lead to collision or malfunction.

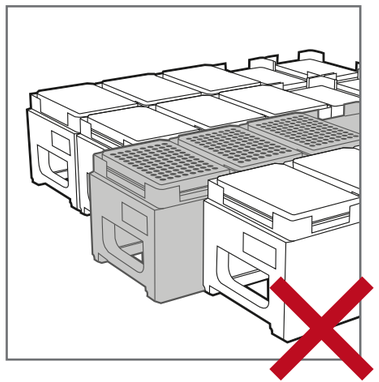


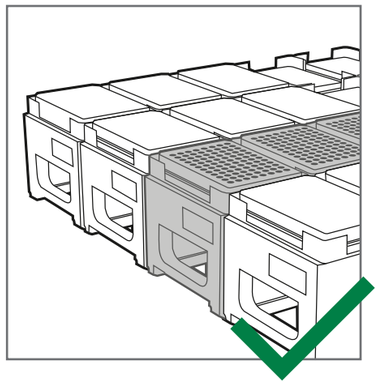
Placing labware

Make sure that labware is placed onto the pedestals correctly.

Incorrectly placed labware will lead to collision or malfunction.

Loading plate and tube carriers

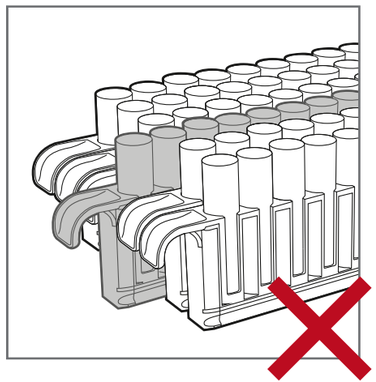


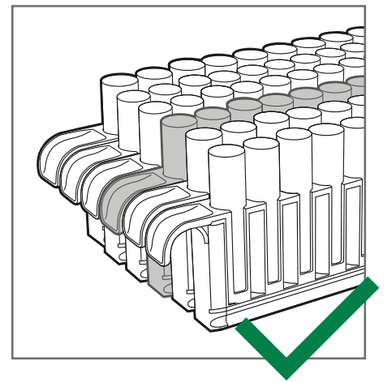


Loading carriers

Make sure the carriers have been pushed to the mechanical stop of the instrument.

Incorrectly positioned carriers will lead to malfunctions or collision.



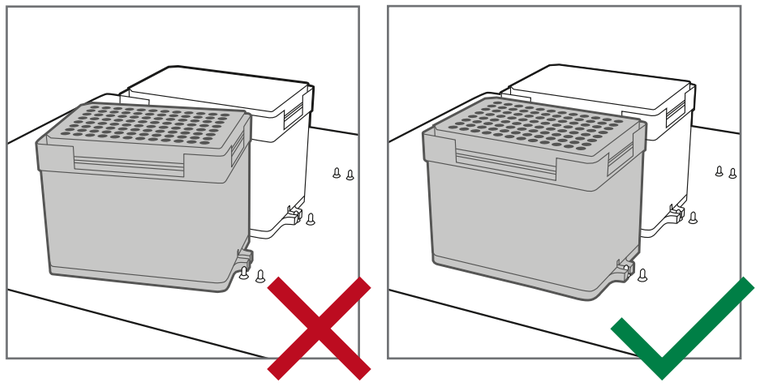


Loading tube carriers

Make sure the tube carriers have been pushed to the mechanical stop of the instrument.

Incorrectly positioned tube carriers will lead to malfunctions or collision.

Placing pedestals

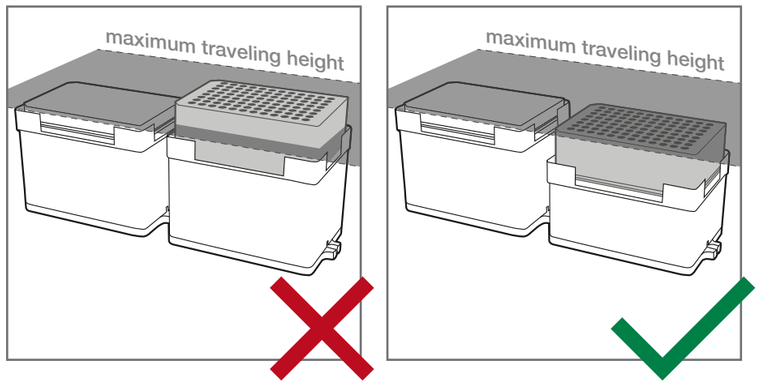


Placing pedestals

Make sure that the pedestals are placed onto the instrument deck correctly.

Incorrectly positioned pedestals will lead to collision or malfunction.

Height of labware on carriers and pedestals

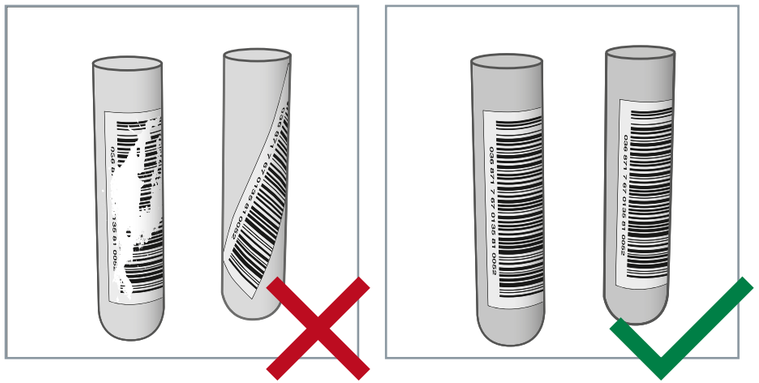


Paring appropriate labware with carrier/pedestal

Make sure not to exceed maximum traveling height.

Make sure to use designated pedestal for deepwell plates.

Barcodes



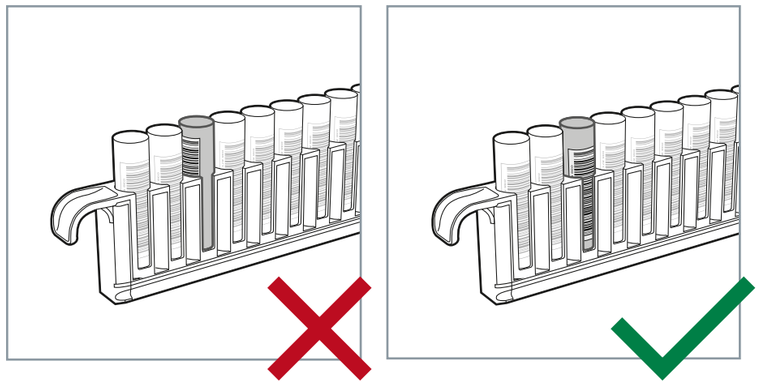
Barcode quality

Make sure that all barcodes are readable as well as correctly applied.

Unreadable, scratched, smudged or incorrectly positioned barcodes on tubes will not be processed.

Refer to the designated instrument Manual for detailed barcode quality requirements.

Barcode orientation

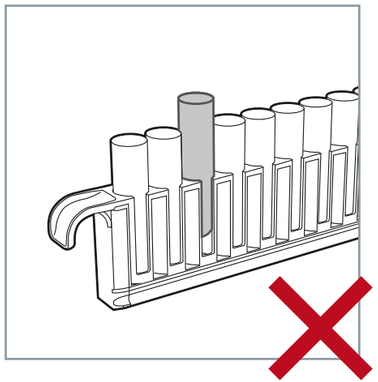


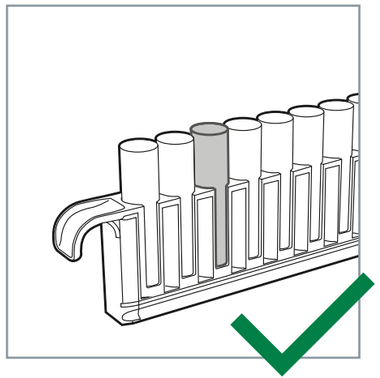
Barcode orientation

Make sure all tube barcodes have the correct orientation.

Incorrectly placed tube barcodes will not be readable.

Placing tubes into tube carriers

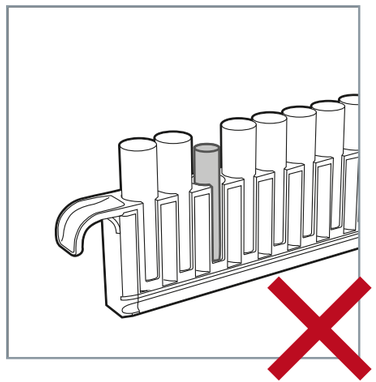


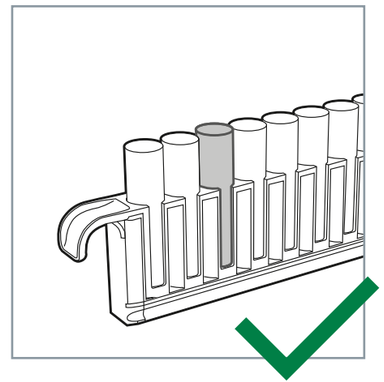


Tube position

Make sure the tubes are pushed all the way down into the tube carrier.

Incorrectly placed tubes may lead to malfunctions.



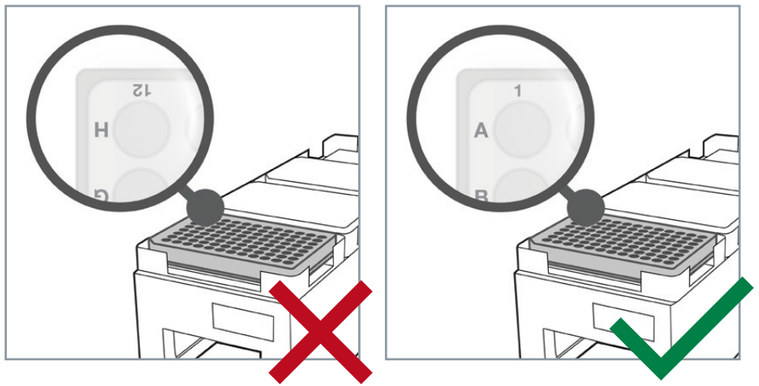


Tube height

Make sure that all tubes placed in the tube carrier are the same height.

Different tube heights will lead to incorrect calculation of sample volumes and lead to malfunctions.

Labware orientation on carriers

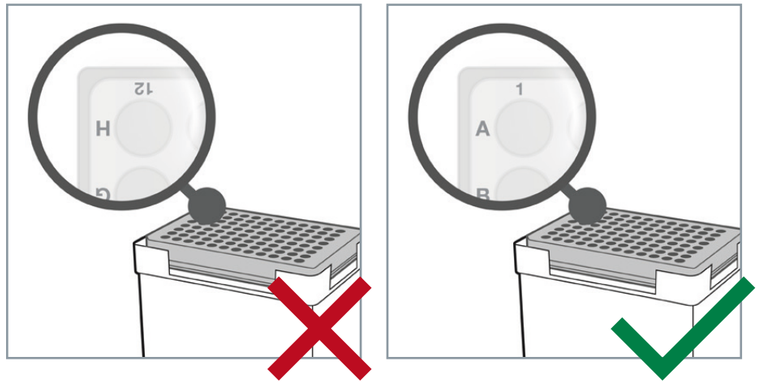


Labware orientation

Make sure plates have the correct orientation on the carrier.

The well position A1 has to be in the upper left.

Labware orientation on pedestals

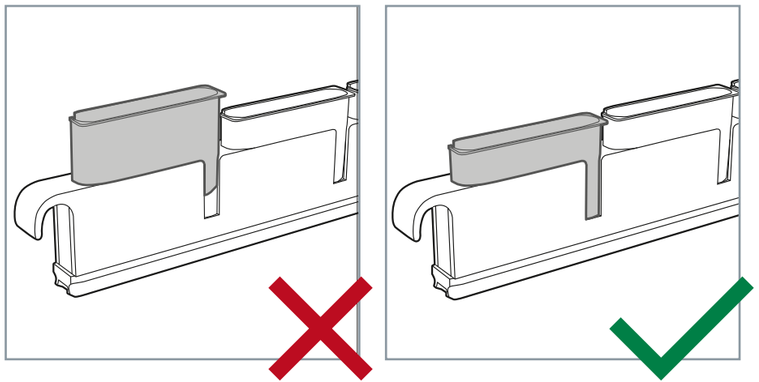


Labware orientation

Make sure plates have the correct orientation on the pedestal.

The well position A1 has to be in the upper left.

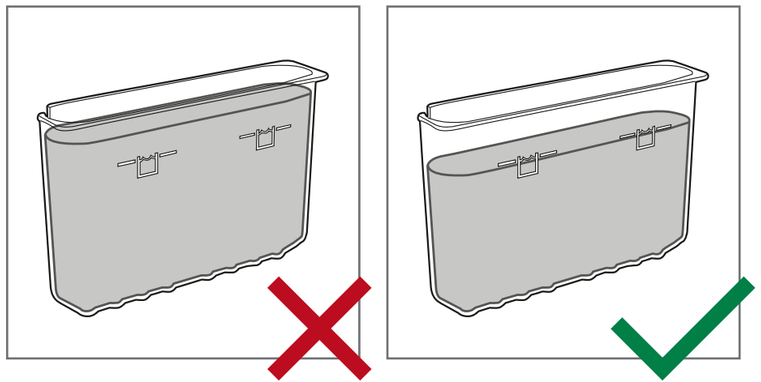
Placing troughs into trough carriers



Trough position

Make sure the troughs are pushed all the way down into the carrier.

Incorrectly placed troughs will lead to spillage and/or malfunctions.

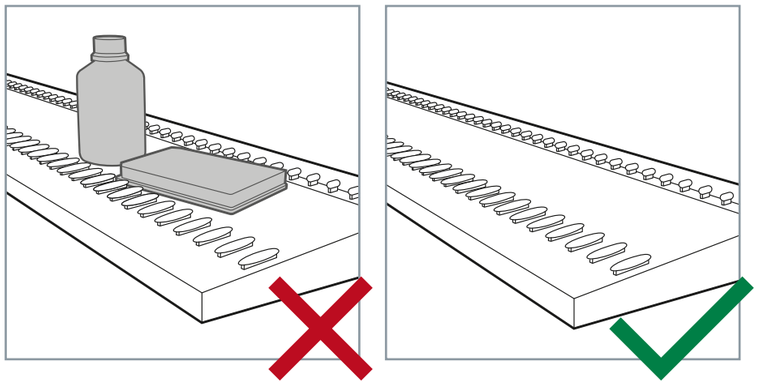


Trough maximum level

Make sure that the troughs are not filled over the maximum level indication mark.

Overfilled troughs will lead to spillage and/or malfunctions.

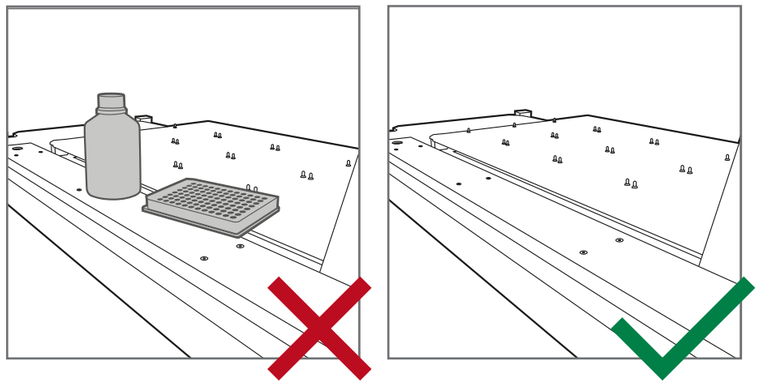
Foreign object damage



Loading tray

Make sure that the loading tray has been cleared from foreign objects before running the instrument.

Foreign objects may obstruct the instrument and lead to damage to equipment.



Instrument deck

Make sure that the instrument deck has been cleared from foreign objects before running the instrument.

Foreign objects may obstruct the instrument and lead to damage to equipment.

Maintenance

Periodic maintenance is required in order to assure safe and reliable operation of your instrument. Daily and weekly maintenance must be performed by the operator in order to keep the instrument clean and to verify proper functioning. Half-yearly maintenance must be performed by a trained HAMILTON technician to maintain proper functioning and verify performance.

HAMILTON recommends that you maintain your instrument at the following intervals:

Intervals

- Daily – Recommended before the instrument is shut-down at the end of the day.
- Weekly – Recommended at the end of the week before the instrument is shut-down.
- Every 6 months – Recommended preventative maintenance by a trained HAMILTON technician.

Decontamination Method

Consult the designated instrument manual for decontamination procedures.

Materials Required

Personal protective equipment (gloves, goggles, protective clothing)

- Lint-free paper towels
- Deionized water
- Disinfectant agent

Daily Maintenance

Daily maintenance requires the following tasks to be performed:

- Inspect the workspace/deck for cleanliness
- Empty the tip waste and liquid waste
- Have the instrument perform a tightness check
- Have the instrument perform a cLLD check

Weekly Maintenance

Weekly maintenance requires the following tasks to be performed:

- Clean the workspace
- Check the carriers and devices for damage
- Empty and clean the solid and liquid waste
- Clean main instrument modules thoroughly
- Clean the instrument covers
- Have the instrument perform a tightness check
- Have the instrument perform a cLLD check

Cleaning

NOTICE
Dilute the concentrated disinfectant agent with deionized water according to the instructions on the bottle.

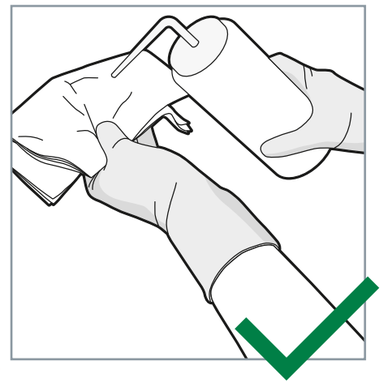
CAUTION
Risk of Damage to equipment.
Do not use cleaning or disinfecting solutions which contain hypochlorite, such as bleach.

CAUTION
Risk of Damage to equipment.
Spraying the instrument or carriers surfaces will lead to damage of equipment.

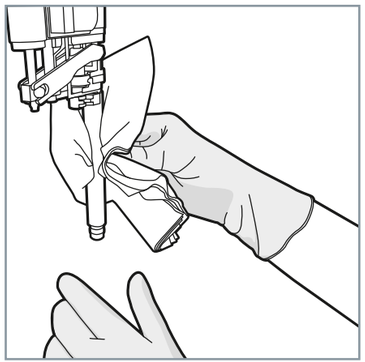
- Do not spray the disinfectant agent onto the instrument or carrier surfaces directly. Spray the liquid onto the paper towel to clean the surfaces and devices.

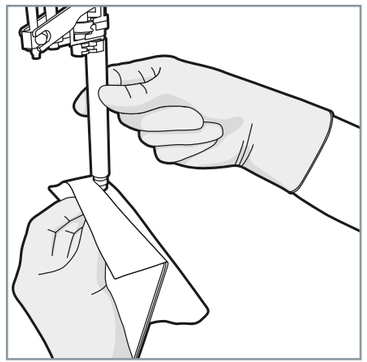






Spray the carriers and instrument parts with disinfectant agent and wipe them clean with paper towels.



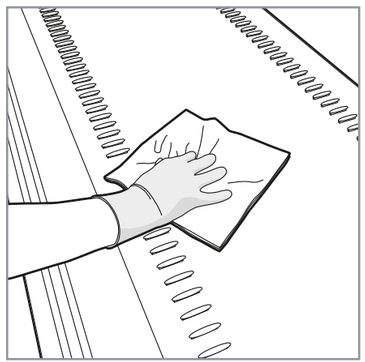


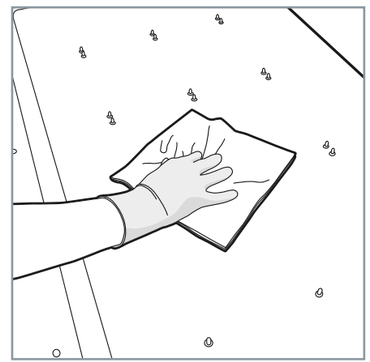
Pipetting Channels

Clean the eject sleeve and tip of the pipetting head with a wetted lint-free paper towel.

Lift the tip eject sleeve and wipe the o-ring and stop disk.

Repeat the procedures for each channel.





Cleaning the Instrument workspace/deck

Spray the instrument workspace/deck with disinfectant agent and wipe with paper towels.

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Good Practice Guide

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